

# Zoravur Singh

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## EDUCATION

Honours Bachelor of Computer Science, Minor in Pure Mathematics

Cumulative GPA: 84%

*University of Waterloo*

2018 — 2024

## SKILLS

**Languages** | Python, Typescript, Go, C/C++, Haskell, Rust, SQL, Lean4, Racket

**Web Development** | React, Redux, WebAssembly, FastAPI, net/http, Express

**Machine Learning** | PyTorch, JAX, NumPy, SciPy, Polars, MLPs, CNNs, Transformers, RL, mechanistic interpretability

**Infrastructure** | PostgreSQL, SQLite, Docker, AWS, Snowflake, CMake, Bun, uv

**Development** | TDD, coding harnesses, prompt engineering, agent orchestration

## EXPERIENCE

### Solutions That Matter

*Toronto, ON*

Software Engineer

Dec 2023 — Present

- Consolidated order entry methods across the company by delivering a custom order processing system end-to-end.
- Saved an estimated ~120 hours/week by reducing processing errors, customer escalations, and status inquiries.
- Developed a reconciliation system for processing compensation reports, saving 150+ hours of work annually.

### DealMaker

*Toronto, ON*

Data Engineering Intern

May 2023 — Aug 2023

- Built analytics dashboards in Metabase using SQL to track sales performance, ad spend ROI, and customer metrics.
- Identified bottlenecks and delivered a 5x performance boost to the ETL service by switching to batch API requests.
- Prototyped a machine learning model to forecast capital raise outcomes using historical transaction data.

### Royal Bank of Canada

*Toronto, ON*

AI Engineering Intern

Sep 2021 — Apr 2022

- Contributed to Aiden, an RL-based trading algorithm handling \$100M+ daily, by architecting new trading signals.
- Developed a C++ market exchange simulation environment, reducing training cycles from 3 days to 14 hours.
- Built MLP classifiers to predict momentum vs. reversion market conditions, improving ETF trading performance.

### Arcturus Networks

*Toronto, ON*

Software Engineering Intern

May 2020 — Aug 2020

- Built a video surveillance application using React, Redux, redux-saga, and ZeroMQ for real-time streaming.
- Implemented live alerts, notifications, and pipeline parameter configuration features.
- Reduced Docker build times from 7 minutes to 20 seconds by optimizing Dockerfiles and CMake cache usage.

### ROSS Intelligence, Inc.

*Toronto, ON*

Machine Intelligence Engineering Intern

May 2019 — Aug 2019

- Extended backend microservices and refactored auth logic to allow users to use the app without creating an account, driving customer acquisition.
- Built a subscription settings page in React with functionality to modify, view, and cancel subscriptions.
- Migrated ~1000 lines of the frontend codebase from JavaScript to TypeScript.

## PROJECTS

**Spock** | A spoken programming language  [zoravur.com/spock-a-spoken](https://zoravur.com/spock-a-spoken)

- Designed and implemented an interpreter for a programming language using English words without punctuation, and integrated OpenAI's speech-to-text model Whisper to enable voice-dictated programming.

**Taan Generator** | Indian classical music generator  (demo)

- Trained a transformer to generate taans (melodic runs) in the Indian classical raga Yaman.